

Updated Anchored Head–Tail Certificate Ledger

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Abstract

We update the exact finite-head/analytic-tail interface after four distinct inputs. RH-252 gives an all-order analytic existence interface for the deterministic target, but its boundary constant M_S is not numerically certified. RH-255 and RH-258 find no anchored head in their two expanded admissibility classes at any of 32 endpoints. RH-259 supplies a finite 23-endpoint quotient block diagnostic whose worst root rate is 0.505642, not a uniform small-noise theorem. We prove the updated gluing estimate and a scoped zero-certificate corollary: the current audited classes have exactly zero complete head–tail certificates. This is a route ledger, not a global nonexistence theorem.

1 The updated certificate interface

Let $\tau_{n,\theta}$ be the residual trace coefficients produced by a candidate cloud or quotient at parameter θ , and let a_n be the deterministic target coefficients of RH-243 [1]. Write

$$L_\theta(z) = - \sum_{n \geq 2} \frac{\tau_{n,\theta}}{n} z^n, \quad L_*(z) = - \sum_{n \geq 2} \frac{a_n}{n} z^n. \quad (1)$$

For a first omitted order $N \geq 3$, define the three nonnegative budgets

$$H_{<N}(\theta, R) = \sum_{n=2}^{N-1} \frac{|\tau_{n,\theta} - a_n| R^n}{n}, \quad (2)$$

$$Q_{\geq N}(\theta, R) = \sum_{n \geq N} \frac{|\tau_{n,\theta}| R^n}{n}, \quad (3)$$

$$A_{\geq N}(R) = \sum_{n \geq N} \frac{|a_n| R^n}{n}. \quad (4)$$

The convention matters: an order-2–12 head has $N = 13$. This first-omitted-order formulation refines the bookkeeping interface of RH-250 [3].

Theorem 1.1 (Budgeted head–tail gluing). *Assume the series in (1) are absolutely convergent on $|z| \leq R$. Then*

$$|L_\theta(z) - L_*(z)| \leq H_{<N}(\theta, R) + Q_{\geq N}(\theta, R) + A_{\geq N}(R). \quad (5)$$

If $H_{<N} \leq h$, $Q_{\geq N} \leq q$, and $A_{\geq N} \leq t$ uniformly on a parameter set, and if $|L_(z)| \leq B_*$ there, then with $D_\theta = e^{L_\theta}$ and $D_* = e^{L_*}$,*

$$|D_\theta(z) - D_*(z)| \leq e^{B_*} (e^{h+q+t} - 1). \quad (6)$$

Consequently a sequence of complete certificates yields locally uniform determinant convergence whenever the three budgets tend uniformly to zero.

Proof. Subtract the two absolutely convergent series. The terms with $2 \leq n < N$ give the finite head $H_{<N}$; the remaining terms are bounded separately by the absolute quotient and target tails. This proves (5). For the determinant, put $\Delta = L_\theta - L_*$. Since $|L_*| \leq B_*$,

$$|e^{L_\theta} - e^{L_*}| = |e^{L_*}| |e^\Delta - 1| \leq e^{B_*} (e^{|\Delta|} - 1),$$

and (6) follows from (5). $\square$

Proposition 1.2 (Analytic target tail exists, but its constant is open). *RH-252 [4] gives a scaled zero-free radius*

$$\rho_* = r_H \lambda = 1.42678748386407 \dots > 1, \quad r_H = 0.85.$$

For every $0 \leq R < S < \rho_*$, the normalized logarithm has a finite boundary supremum M_S and

$$A_{\geq N}(R) \leq M_S \frac{(R/S)^N}{1 - R/S}. \quad (7)$$

At $R = 1$, $N = 13$, and $S = 1.35$, the geometric factor per M_S is 0.07796985628233917. The existence of M_S is exact; no certified numerical upper bound for it is available in the archived audit.

Proof. This is the Cauchy estimate for the holomorphic logarithm on the disk supplied by RH-252. The coefficient of z^n is $-a_n/n$, so summing the resulting geometric series gives (7). The final numerical factor is only the displayed geometric factor; multiplying it by a truncated boundary scan would not certify M_S . $\square$

2 Scoped zero-certificate result

The head classes are deliberately kept separate. RH-255 uses a convex single-use shell box containing every prefix and binary shell subset. Its archived primal–dual audit has 0 passes at all 32 endpoints. RH-258 tests the unit-cap signed-integer lattice $\{-1, 0, 1\}^J$, with exact-integrality MILP solutions and zero reported MIP gap; it also has 0 passes at all 32 endpoints. Neither audit supplies an operator realization for an arbitrary mask [5, 6].

Corollary 2.1 (Zero complete certificates in the audited classes). *Restrict the candidate head to the union of the RH-255 box class and the RH-258 unit-cap signed-integer class, with the archived 32-endpoint tolerance protocol. There is no complete head–tail certificate in this restricted class.*

Proof. Every complete certificate has, as a necessary first component, a passing anchored finite head. The RH-255 audit supplies 0/32 such passes and the RH-258 audit supplies 0/32. Therefore their union supplies no passing head, so the set of complete certificates is empty, independently of the tail budgets. This conclusion is scoped to the two archived classes and their finite tolerance protocol. $\square$

The two classes cover 64 class–endpoint cases. The RH-255 audit excludes

$$62,030,604,700$$

eligible binary subsets through the box obstruction. RH-258’s unit-cap lattice contains

$$39,417,456,084,975,216$$

signed points in aggregate. These counts are finite audit descriptors, not claims about all possible selectors.

3 Quotient-tail ledger

RH-259 extends the ordered-Schur quotient calculation to dimension 1024 [2, 7]. All 23 audited twelfth powers are contractive, but the worst root rate is

$$q_{12} = 0.5056418005507071,$$

at the right channel with $\sigma = 0.0025$ and dimension 1024. The finite unit-disk logarithmic tail diagnostic is $5.654507945432548 \times 10^{-4}$. Nine archived endpoints remain outside the calculation, and the floating matrix norms are not interval enclosures. Thus this is a family of finite diagnostics, not a uniform small-noise quotient certificate.

certificate component	archived status
deterministic target analytic tail	exact existence; M_S not certified
RH-255 expanded single-use head	0/32 passes
RH-258 unit-cap signed-integer head	0/32 passes
RH-259 quotient block tail	23 finite endpoints; uniformity open
cloud coefficient bridge	open
complete head–tail certificate	0

4 Protocol, boundary, and next input

The experiment reads the four archived JSON ledgers, checks 44 source consistency predicates, and writes a new structured ledger. It does not rerun the dimension-1024 Schur calculation or fit new moments. The result therefore separates exact inherited interfaces from finite numerical audits: the target-tail existence flag is true, while a certified boundary constant, a legal head pass, a cloud coefficient bridge, and a uniform quotient tail are all false.

The obstruction does not exclude larger integer caps, signed or complex selectors derived from an invariant quotient, larger resolved windows, a future interval bound for M_S , or another operator realization. It also does not establish global nonexistence. Gates A–E remain false/open. No Hilbert–Polya operator is constructed; no Riemann zeros are identified; no equality with the completed zeta divisor and no implication of RH is claimed.

The next reopening input must supply at least one genuinely new legal head, a certified M_S , a uniform quotient theorem, or an exact coefficient bridge; the useful route requires the relevant obligations together rather than a finite-order extrapolation.

References

- [1] Bin Wang. A deterministic numerator coefficient-anchor dictionary. *Preprint*, 2026. RH-243 preprint.
- [2] Bin Wang. Block-power quotient envelope criterion. *Preprint*, 2026. RH-246 preprint.
- [3] Bin Wang. Anchored head–analytic tail gluing criterion. *Preprint*, 2026. RH-250 preprint.
- [4] Bin Wang. Deterministic-numerator analytic-tail certificate. *Preprint*, 2026. RH-252 preprint.
- [5] Bin Wang. Expanded-window anchored zonotope obstruction. *Preprint*, 2026. RH-255 preprint.
- [6] Bin Wang. Unit-cap signed-integer selector obstruction. *Preprint*, 2026. RH-258 preprint.
- [7] Bin Wang. Extended quotient block-power diagnostic. *Preprint*, 2026. RH-259 preprint.